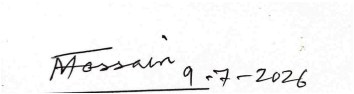




Daffodil International University
Department of Computer Science and Engineering
Course Code: CSE499 Title: FYDP (Title Defense)

Proposal Approval Form

I. 1st Member Information			
Student ID:	232-15-269	Student Name:	Aupurba Sarker
II. 2nd Member Information			
Student ID:	232-15-473	Student Name:	Labony Sur
Semester:	Summer 2026	Date:	09/07/2026
III. Proposed Title:	An Explainable Knowledge Graph-Based Framework for Personalized Drug Repurposing for Bangladeshi Lung Cancer Patients		
IV. Updated Title (If there is any update):			
This is the approval for the Final Year Design Project (FYDP) proposal and title submitted by the above-mentioned student who is from the Department of CSE under my supervision. Upon thorough review and evaluation, it is my professional opinion that the proposed project title [mentioned in Sec III and IV] is both relevant and aligned with the academic goals and standards of our department. I (Supervisor) have endorsed the proposed proposal and title. Under our department's guidelines, the student has fulfilled all the necessary prerequisites and will attach this Proposal Approval Form to their FYDP Title Phase Evaluation Report.			
Supervisor Name:	Dr. Md. Fokhray Hossain		
Supervisor Designation:	Dean & Professor		
<u>Approved by:</u>			
			
_____ Supervisor Signature with Date			

V. Detailed Proposal

1. Objective

The primary objective of this research is to design, implement, and evaluate an explainable artificial intelligence (XAI) framework based on a heterogeneous biomedical knowledge graph to identify personalized, repurposed drug candidates for lung cancer patients in Bangladesh. This initiative directly addresses the critical national need for accessible, targeted cancer therapies while advancing the global application of interpretable AI in precision medicine.

Traditional drug discovery is hindered by immense financial costs, high attrition rates, and extended development timelines that often span over a decade. While computational drug repurposing offers a significantly faster and more cost-effective alternative by identifying new therapeutic indications for existing, regulatory-approved pharmaceuticals, current AI-driven models face two major barriers:

1. **The "Black Box" Problem:** Most deep learning architectures generate drug predictions without providing transparent biological reasoning. Without readable underlying mechanistic evidence, clinicians cannot evaluate the safety or biological plausibility of the predictions, severely hindering real-world adoption in oncology.
2. **Lack of Population Localization:** Current precision oncology tools rely almost exclusively on Western genomic cohorts. They fail to capture the distinct somatic mutation frequencies, genomic variations, and demographic profiles of Bangladeshi lung cancer patients, leading to potential treatment bias.

To overcome these structural and clinical challenges, the proposed framework will fulfill the following core operational objectives:

1.1. Construct a Localized Biomedical Knowledge Graph:

Aggregate and standardize primary clinical records and somatic mutation profiles from Bangladeshi lung cancer patients, systematically integrating them with comprehensive global biomedical repositories, including DrugBank, Open Targets, STRING, Reactome, and DGIdb. This will yield a multi-layered, heterogeneous knowledge graph mapping complex interactions across patients, diseases, target genes, proteins, pathways, and pharmacological agents.

1.2. Implement Graph Neural Networks (GNNs) for Link Prediction:

Develop and train advanced Graph Neural Network models, such as Relational Graph Convolutional Networks (R-GCNs), to learn low-dimensional graph embeddings and effectively capture non-linear topological interactions within the biomedical network. These models will perform link prediction tasks to accurately identify unannotated, high-affinity drug-disease associations tailored to individual patient mutation profiles.

1.3. Extract Explainable Biological Pathways via XAI:

Integrate path-based explainable AI algorithms to traverse the knowledge graph and automatically extract human-readable molecular traversal paths. Instead of merely outputting a ranked list of candidate drugs, the framework will explicitly detail the specific gene targets, protein-protein

interactions, and biological pathways driving each prediction, ensuring full mechanistic transparency.

1.4. Deliver a Clinical Decision- Support System:

Benchmark and validate the predicted drug candidates against established drug-response datasets and peer-reviewed biological literature. The final output will serve as a localized, evidence-backed clinical decision-support tool that bridges the gap between complex AI predictions and practical oncological practice, enabling medical professionals in Bangladesh to make trusted, personalized, and data-driven treatment decisions.

2. Motivation

The primary motivation for this research stems from the urgent need for accessible, effective, and individualized treatment options for lung cancer patients in Bangladesh. Lung cancer remains one of the most prominent health challenges in the country, where high treatment costs and limited healthcare infrastructure restrict access to targeted therapies. Traditional drug discovery is an exceptionally slow and expensive process, often requiring over a decade and billions of dollars to bring a single new medicine to market. Computational drug repurposing offers a practical alternative by identifying novel therapeutic uses for existing, safety-approved drugs, thereby significantly reducing both development time and cost.

Despite the clear benefits of computational drug discovery, existing artificial intelligence models in precision oncology face two major limitations that prevent their effective adoption in local clinical settings:

2.1. Genomic Bias and Lack of Population Localization:

The vast majority of current biomedical datasets and artificial intelligence models are trained exclusively on Western populations. Consequently, these models overlook the distinct genetic variations, somatic mutation frequencies, and unique environmental factors present in Bangladeshi patients. Applying Western-trained models directly to local patients can lead to inaccurate predictions and biased therapeutic recommendations.

2.2. The Black Box Problem in AI:

Deep learning models used for drug prediction generally function as opaque systems. They output prospective drug candidates without providing the underlying biological rationale or showing which molecular pathways are involved. In clinical oncology, doctors cannot safely prescribe treatments based solely on unexplained computer outputs. Clinicians require clear, verifiable biological evidence to trust and act upon AI-generated advice.

This research directly addresses these fundamental challenges by building an explainable knowledge graph framework specifically localized with Bangladeshi patient data. By integrating primary clinical profiles with global pharmacological databases and using explainable artificial intelligence to highlight readable biological pathways, this project aims to provide local oncologists with a transparent, cost-effective, and trustworthy clinical decision support tool.

3. Background Study

The evolution of precision oncology has transformed modern cancer treatment by shifting the focus from generalized chemotherapy to targeted therapies based on individual genetic and molecular profiles [1]. To overcome the massive financial burdens and decade-long timelines associated with traditional drug discovery, computational drug repurposing has emerged as a vital strategy [2]. This approach uses advanced computational methods, structural dynamics, and virtual screening to identify new therapeutic indications for existing, regulatory-approved pharmaceuticals [2], [10]. Recent *in silico* investigations on lung cancer, such as Non-Small Cell Lung Cancer (NSCLC) targets, further demonstrate the effectiveness of repurposing FDA-approved candidates to bypass early preclinical hurdles [10].

In recent years, biomedical Knowledge Graphs (KGs) have become a foundational tool for computational drug repurposing [2]. Knowledge graphs represent biological concepts such as drugs, genes, proteins, biological pathways, and diseases as nodes, while representing their biological relationships as directed or undirected edges [3], [4]. By organizing vast amounts of heterogeneous data into a single unified network, knowledge graphs allow researchers to model complex biological systems comprehensively [3].

To analyze these complex biological networks, deep learning techniques, particularly Graph Neural Networks (GNNs), have gained widespread adoption [3], [4]. Architectures like Relational Graph Convolutional Networks (R-GCNs) and geometric GNNs learn low-dimensional embeddings that capture non-linear topological interactions across interconnected biological entities [3], [5], [9]. Through link prediction tasks and cross-attention mechanisms, GNNs can predict unannotated drug-disease connections, evaluate drug-target interaction affinity, or identify key cancer driver genes for therapeutic intervention [3], [4], [9].

However, traditional deep learning architectures often function as opaque black boxes, generating candidate predictions without disclosing the underlying biological reasoning [2], [5]. To address this lack of interpretability, recent research has incorporated Explainable Artificial Intelligence (XAI) and path-based reasoning algorithms into graph-based learning [2], [5], [9]. These methods traverse the network to extract human-readable molecular paths and cross-attention weights, revealing intervening targets to justify computational predictions medically [2], [5], [9]. Furthermore, integrated machine learning frameworks have also been utilized to model survival outcomes and overcome treatment resistance in multicenter lung cancer cohorts [11].

Despite these technological advancements, significant literature gaps remain in current precision oncology frameworks. First, existing knowledge graph models depend almost exclusively on Western clinical and multi-omics datasets [6]. As a result, they fail to capture the distinct somatic mutation patterns, environmental exposures, and genomic characteristics of lung cancer patients in Bangladesh [6]. Second, many current explainable AI outputs produce complex network features that remain difficult for clinicians to interpret quickly in resource-constrained medical environments [7]. Finally, most drug repurposing models focus on general disease indications rather than generating individualized, patient-specific treatment plans based on a patient's unique mutation profile [8]. This research directly addresses these limitations by developing a localized,

explainable knowledge graph framework specifically tailored for Bangladeshi lung cancer patients [1], [6].

A comprehensive summary of the literature review, detailing the datasets, architectures, performance metrics, limitations, and corresponding reference citations, is presented in Table 3.1.

TABLE 3.1 Comprehensive Summary of Literature Review for Drug Repurposing and Precision Oncology

Author & Reference	Data / Dataset Used	Dataset Format	Model / Algorithm Used	Accuracy / Performance	Key Limitations
Ryu et al. (2026) [1]	OncoKB, PubMed, TxGNN KG, ClinicalTrials.gov, CPTAC	Tabular, MAF files, Graph Triples, API Streams	Rule-based Heuristic Prioritization Pipeline (IDAP)	Expanded evidence hypotheses; recovered matched CIViC therapy in 28/41 samples	Literature module relies on co-mention (non-causal); heuristic scores sensitive to weight choices; lacks deep XAI path extraction.
Perdomo-Quintero et al. (2026) [2]	NeDRex Knowledge Graph, DrugMechDB	Heterogeneous Graph (Proteins, Genes, Drugs, Diseases)	Heterogeneous GraphSAGE + XAIPath (MinHash & K-means Clustering)	AUROC > 0.95, AUPRC > 0.90, Precision/Sensitivity > 85%, Hits@1 = 0.71	Simple path evaluation exponentially increases computation time; relies on existing KG knowledge without patient-level multi-omics.

Khodadi AghGhal eh et al. (2026) [3]	RepoDB, SIDER, DrugBank, STRING, DisGeNET, RDKit	Labeled Heterogeneous Directed Multigraph, Morgan Fingerprints	Word2Vec + Dual-Channel 1D-CNN (IDC_Conv 1D) with WBCE Loss	AUC-ROC = 0.9836, AUC-PR = 0.9686, Accuracy = 95.73%, F1 = 0.9074	Optimization relied on trial-and-error or grid search; lacks an XAI module for human-readable biological pathway extraction.
Li & Xie (2026) [4]	TCGA, NCG, COSMIC, PPI, RegNetwork, DawnNet, KEGG	Pan-cancer Multi-omics Matrix (Mutation, Methylation, Expression) + 6 Multiplex Networks	MNDGNN (Multiplex Networks-based Directed Graph Neural Network)	AUROC = 0.8780, AUPRC = 0.8962, F1-Score = 0.8238	Equal weighting across multiplex networks; noisy edges affect performance; identifies driver genes rather than drug repurposing candidates.
Gonzalez-Cavazos et al. (2026) [5]	MIND Knowledge Graph, DrugMechDB	Heterogeneous Biomedical Knowledge Graph	DBR-X (Case-Based Reasoning + R-GCN + Path-Enforcing Mask)	MRR = 0.3770, Hits@1 = 0.2796, Hits@10 = 0.5708	Similarity retrieval relies on simple inner products (ignores multi-hop dependencies); lacks localized genomic profiling.

Islam et al. (2023) [6]	NICRH Hospital Cancer Registry, Bangladesh Patient Cohort	Retrospective Clinical Records, Survey Data	Statistical Survival Analysis, Cox Proportional Hazards Model	Characterized demographic differentials, stage distribution, and treatment landscape in Bangladesh	Observational clinical dataset; lacks AI/GNN computational modeling or drug repurposing integration.
Aamer et al. (2026) [7]	PrimeKG (17,080 diseases, 4M+ relations), Recent FDA Approvals	Multimodal KG Triples, Random-Walk Sampled Paths	COMIC (Contrastive Masking with Feature & Path Selection Branches)	AUPRC = 0.88–0.96, 9.55% average gain over TxGNN, 70% prediction on new FDA pairs	Performance degrades on sparse disease graphs; drug-drug interaction congestion obscures mechanistic paths.
Abo-Dahab et al. (2026) [8]	ChEMBL 36 (3,127 drugs, 1,156 proteins, 1,065 indications)	Tri-modal Knowledge Graph, ESM-2 Protein Embeddings	TransR, ComplEx, DistMult, Heterogeneous GNN (GraphSAGE / Blackwell GNN)	GNN ROC-AUC = 0.9762 (Drug-Protein) / 0.9881 (Drug-Indication), PR-AUC = 0.5785 / 0.8060	Relies heavily on network topology; transductive setup limits performance on unobserved cold-start entities without topological context.
Xiong et al. (2025) [9]	BioSNAP, DrugBank, BindingDB, KIBA	Graph Network Triples, Protein Sequences & Structures	Geometric Graph Neural Network (GPS-DTI) with Cross-Attention Mechanism (CAM)	AUROC = 0.932, AUPRC = 0.915, High Out-of-Distribution Generalizability	Focuses on paired drug-target binding affinity rather than complex multi-layered patient

					Knowledge Graphs.
Khan et al. (2025) [10]	DrugBank (3,500 FDA-approved drugs), RCSB PDB (MEK1 structure)	3D Structural Coordinates, PDBQT Files, Molecular Trajectories	Virtual Screening, Molecular Docking & 100 ns Molecular Dynamics (MD) Simulations	High binding affinity (Radotinib: -10.8 kcal/mol, Alectinib: -10.6 kcal/mol)	In silico structural approach; lacks population-specific patient multi-omics or GNN-based link prediction.
Wu et al. (2025) [11]	Multicenter NSCLC Cohort, Aumolertinib Resistance Profiles	Clinical Survival Records, Drug Sensitivity Matrices	Integrated Machine Learning Survival Framework	High consensus predictive performance for drug resistance candidate screening	Focuses on machine learning survival modeling; lacks explicit XAI graph path extraction.

4. Gap Analysis

Although existing computational research has made notable progress in artificial intelligence, molecular dynamics, and precision drug discovery, several critical gaps prevent these technologies from being safely deployed in real-world clinical settings [1], [2], [10], [11]. Current state-of-the-art models suffer from three fundamental limitations: geographical data bias, a lack of model interpretability, and insufficient patient-level personalization [5], [6], [8], [9].

First, nearly all existing knowledge graph and virtual screening frameworks rely exclusively on Western clinical, structural, and genomic datasets [6], [10]. Because genetic profiles, somatic mutation patterns, and environmental exposures vary significantly across populations, these global models fail to capture the specific disease characteristics of lung cancer patients in Bangladesh [6]. Second, standard deep learning architectures and molecular modeling techniques often act as opaque black boxes or rely on complex mathematical spaces, providing candidate drug predictions without revealing human-readable biological pathways [5], [7], [9]. Without clear medical rationale, oncologists cannot evaluate the safety or biological plausibility of a predicted drug [5], [9]. Finally, most existing platforms focus on broad drug-disease or drug-target interactions rather than delivering individualized recommendations derived from a specific patient's multi-omics profile [8], [9], [11].

The specific research gaps alongside the localized solutions provided by this project are summarized in Table 2.

TABLE 2 Summary of Identified Literature Gaps and Corresponding Proposed Solutions

Gap Category	Current Literature Limitations	Proposed Framework Solutions
Geographical & Demographic Representation	Models and virtual screens rely almost entirely on Western genomic, structural, and clinical datasets, leading to population bias [6], [10].	Integrates primary clinical and somatic mutation data from Bangladeshi lung cancer patients [1], [6].
Model Interpretability & Trust	Deep learning and interaction models operate as black boxes or lack intuitive path-level explanations [5], [7], [9].	Uses path-based Explainable AI (XAI) and attention weights to extract clear, human-readable molecular paths [2], [5], [9].
Level of Personalization	Models predict general drug-disease links or target affinity without tailoring to individual mutation profiles [8], [9], [11].	Generates personalized drug repurposing candidates based on specific patient mutation data [1], [8].

By addressing these core limitations, this project bridges the gap between global computational frameworks and local clinical application [1], [6]. The proposed framework provides local oncologists with a localized, transparent, and evidence-backed decision-support tool tailored to the needs of Bangladeshi lung cancer patients [1], [6].

5. Methodology

This project will be executed by developing a localized biomedical knowledge graph that integrates public pharmacological data with specific clinical and genomic profiles of Bangladeshi lung cancer patients. The approach involves representing biological entities like genes, proteins, and diseases as nodes, and their interactions as edges. A Graph Neural Network will be trained on this complex graph to predict new therapeutic uses for existing medications through a process called "link prediction," while an explainable artificial intelligence module will concurrently extract clear biological pathways to logically justify these recommendations.

5.1. Techniques/Tools:

1. **Graph Neural Networks:** Advanced deep learning models, such as Relational Graph Convolutional Networks, will be used to learn and map the complex connections within the biological networks.
2. **Explainable AI:** Path-based extraction techniques will be applied to generate transparent medical rationales instead of traditional black-box outputs.
3. **Software Tools:** The system will be built using Python, specifically utilizing the PyTorch library for deep learning and the NetworkX library for graph construction.

5.2. Data Collection:

1. Primary clinical and somatic mutation data will be collected from Bangladeshi hospitals or research institutions after obtaining the required permissions.
2. Public biomedical data will be collected from trusted databases such as DrugBank, Open Targets, STRING, Reactome, and DGIdb.
3. All datasets will be cleaned, integrated, and standardized before constructing the knowledge graph.

5.3. Data Analysis:

1. The collected datasets will be analyzed to identify relationships between patients, genes, mutations, drugs, and biological pathways.
2. The framework will recommend personalized drug candidates based on each patient's mutation profile.
3. The performance of the framework will be evaluated using Precision, Recall, F1-score, and ROC-AUC.
4. The recommended drugs will also be validated by comparing them with existing biomedical evidence and published literature.

5.4. Workflow Diagram:

The end-to-end computational pipeline for drug repurposing begins with parallel data acquisition, combining primary Bangladeshi clinical and somatic mutation data with public biomedical databases, including DrugBank, Open Targets, STRING, Reactome, and DGIdb, as illustrated in Fig. 1. The collected datasets undergo cleaning, preprocessing, identifier mapping, and standardization before integration into a unified format. Subsequently, a heterogeneous biomedical knowledge graph is constructed to extract features via graph embeddings. Link prediction models are then applied to these embeddings to identify personalized drug candidates and extract knowledge graph paths for explainable recommendations. Finally, the candidate predictions are validated against drug response data and published scientific literature to yield final personalized drug recommendations.

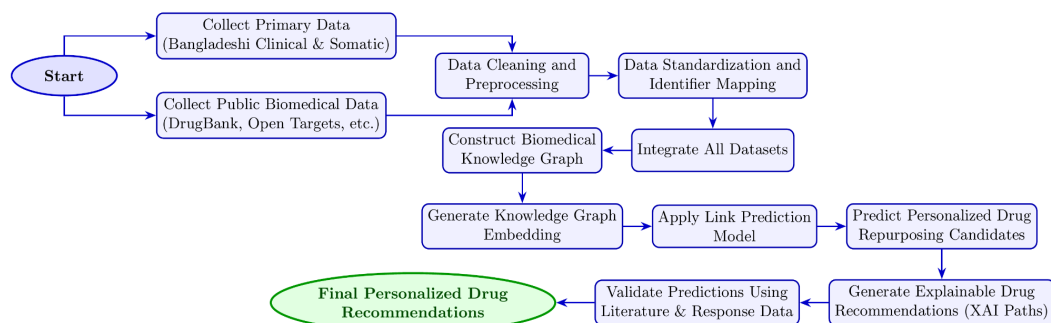


Figure 1. Overall computational workflow for knowledge graph-based personalized drug repurposing using clinical and public biomedical datasets.

6. Expected Outcomes

Upon successful execution, this research will produce several key scientific artifacts, practical clinical tools, and significant economic advantages for the healthcare sector in Bangladesh:

6.1. Localized Biomedical Knowledge Graph:

The successful architecture and construction of a multi-layered, heterogeneous biomedical knowledge graph that integrates primary clinical parameters and somatic mutation profiles specifically collected from Bangladeshi lung cancer patients. This localized graph will systematically unify patient-specific genomic variations with comprehensive global biomedical repositories, including DrugBank, Open Targets, STRING, Reactome, and DGIdb, creating a robust topological mapping of complex gene, protein, disease, and drug interactions.

6.2. Personalized AI Prediction Engine:

The development of an advanced computational prediction framework driven by Graph Neural Networks (GNNs), such as Relational Graph Convolutional Networks (R-GCNs). By leveraging low-dimensional graph embeddings and link prediction algorithms, this engine will evaluate individual somatic mutation signatures to predict novel, highly targeted drug repurposing candidates tailored to the distinct genomic profile of each patient.

6.3. Mechanistic Explainable AI Module:

The implementation of an interpretable path-extraction XAI module designed to eliminate the opaque "black-box" nature of traditional deep learning architectures. This module will dynamically extract and visualize clear, human-readable molecular traversal paths across the knowledge graph, presenting explicit biological evidence, including intervening target genes, protein interactions, and cellular pathways, to medically justify every drug recommendation.

6.4. Clinically Validated Therapeutic Candidates:

A systematically prioritized and empirically benchmarked catalog of cost-effective, personalized drug repurposing candidates. Predictions will undergo rigorous validation against established

drug-response benchmarks, cellular sensitivity data, and peer-reviewed biological literature to ensure therapeutic safety and mechanistic viability.

6.5. Clinical Decision Support System:

The delivery of a practical, transparent, and evidence-backed clinical decision-support framework tailored for oncologists in Bangladesh. By translating complex graph neural network outputs into intuitive biological pathways and actionable recommendations, this system bridges the gap between advanced artificial intelligence and daily clinical oncology, enabling medical professionals to make trusted, personalized, and data-driven treatment choices.

6.6. Strategic Reduction in R&D and Drug Development Costs:

A substantial economic optimization of the pharmaceutical discovery pipeline. By computationally identifying off-label applications for existing, regulatory-approved pharmaceuticals whose safety and pharmacokinetic profiles are already established, this framework circumvents the immense financial capital, high failure rates, and decade-long timelines typically required for De Novo drug discovery and early-stage preclinical trials.

6.7. Financial Relief for Cancer Patients and Healthcare Infrastructure:

The generation of significant economic benefits for both individual patients and broader public health institutions in resource-constrained environments like Bangladesh. Identifying affordable, locally available alternative medications directly reduces out-of-pocket prescription expenses for cancer patients while simultaneously decreasing overall financial burdens, import dependencies, and resource constraints on national healthcare systems and hospitals.

7. Conclusion

This research proposes an explainable, knowledge graph-driven artificial intelligence framework designed specifically for personalized drug repurposing in Bangladeshi lung cancer patients. Computational drug repurposing provides a faster, significantly more cost-effective alternative to traditional drug discovery, offering viable treatment strategies in resource-constrained environments where access to expensive targeted therapies remains limited.

The primary contribution of this work lies in its localized, population-specific design and its emphasis on model interpretability. By integrating primary Bangladeshi clinical records and somatic mutation data with global biomedical repositories, the framework eliminates population bias and ensures that recommendations are genetically relevant to local cohorts. Furthermore, by incorporating path-based explainable artificial intelligence, the system replaces opaque "black box" deep learning predictions with transparent, human-readable molecular pathways.

In addition to its clinical value, the framework offers substantial economic benefits by reducing pharmaceutical research costs and lowering the financial burden on patients and local healthcare systems. Ultimately, this project delivers a reliable, transparent, and evidence-backed clinical decision-support tool that bridges computational AI models with practical oncological decision-making in Bangladesh.

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